



YOLOV5 파인튜닝

XPU 기반 학습 설정 가이드



Run 설정

학습 명령어 및 파라미터

```
bash

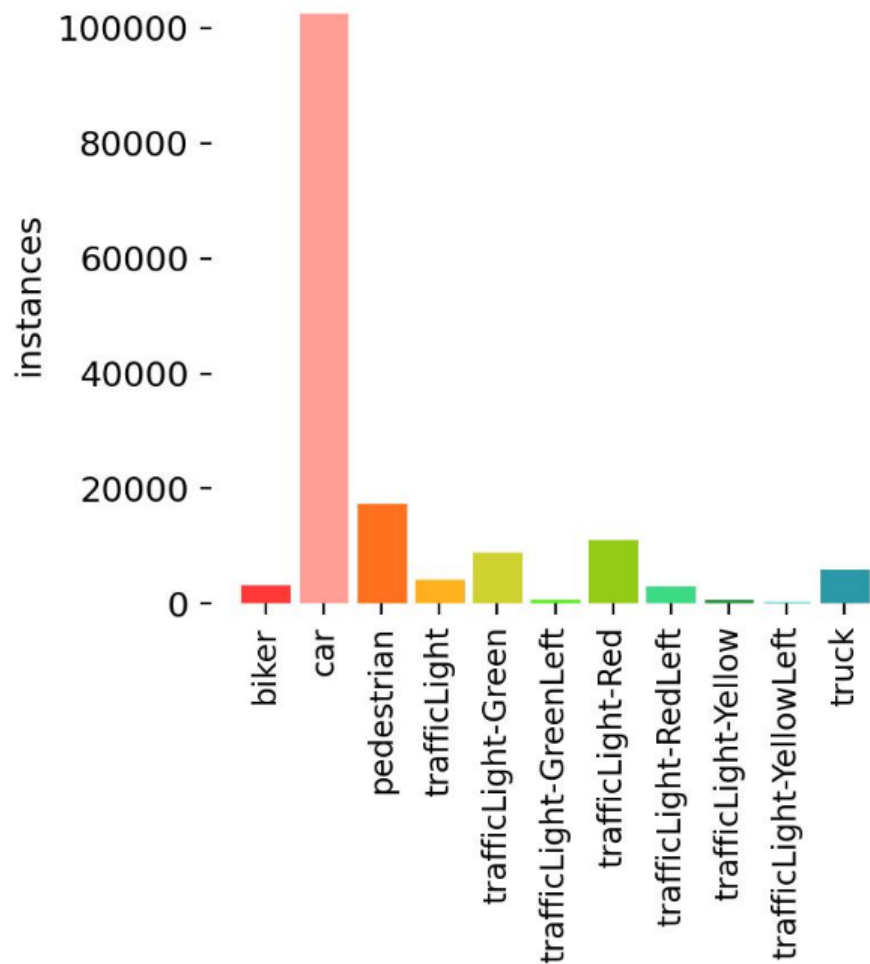
python train_xpu.py \
  --data ".../Self Driving Car.v3-fixed-small.yolov5pytorch/data.yaml" \
  --weights yolov5m.pt \
  --epochs 10 --patience 0 \
  --img 640 --batch 20 \
  --name udacity_yolov5m_xpu
```

• 디바이스: Intel GPU(XPU) / dataloader workers=8

- Backbone/Head: pretrained from yolov5m.pt
- Loss: Box(0.05) + Obj(1.0) + Cls(0.5)
- Opt & LR: SGD (lr0=0.01, mom=0.937, wd=5e-4) + Linear schedule, warmup 3ep
- Aug: HSV(0.015/0.7/0.4), translate 0.1, scale 0.5, fliplr 0.5, mosaic ON
- Tricks: EMA, grad-clip 10, AutoAnchor, EarlyStop(X patience=0)
- I/O: 640×640, batch 20, epochs 10, XPU



데이터셋 소개



Udacity Self-Driving Car (Roboflow "fixed-small v3")

Overview

목적: 자율주행 도로 주행 장면에서
다중 객체 탐지

출처: Roboflow 재라벨링 fixed 버
전의 512×512 축소본(small)

규모: 15,000 images, 97,942
labels, 11 classes

라이선스: MIT License

특징: 원본 Udacity의 라벨 누락·오
류를 수정해 배포된 버전

Class

car, truck, pedestrian, biker

trafficLight 계열 7종
trafficLight, Red, Yellow, Green,
RedLeft, YellowLeft, GreenLeft

소물체(신호등) 비중을 높이기 위해
→ 입력 해상도 640 사용(작은 객체
인식에 유리)

클래스 불균형이 심하다

Issue

연속 프레임
같은 이미지 2장씩 중복되어있음

중복 박스(duplicate bbox) 일부
→ $IoU \geq 0.95$ 중복 자동 제거

Null 이미지(라벨 없음) 포함
→ 학습/평가에서 자동 무시

Split: 80/10/10(scene 누수 방지),
결과 train/val/test 경로로 구성

클래스별 결과

강한 클래스:

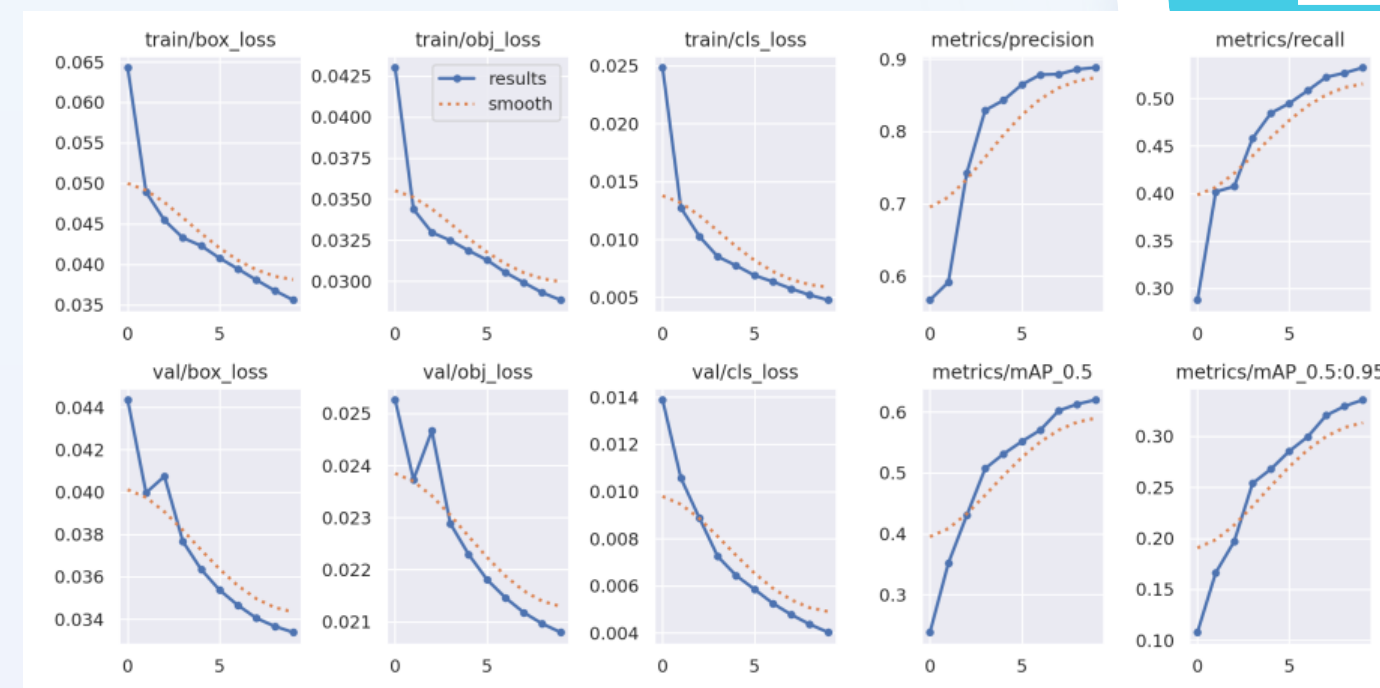
car(mAP@50 0.87, R 0.82),
truck(0.85, 0.80),
trafficLight(0.841, 0.779),
trafficLight-Red(0.847, 0.771),
trafficLight-RedLeft(0.848, 0.74)

약한 클래스:

trafficLight-GreenLeft(mAP@50 0.36, R≈0),
trafficLight-Yellow(0.047, R≈0),
trafficLight-YellowLeft(≈0, R=0)

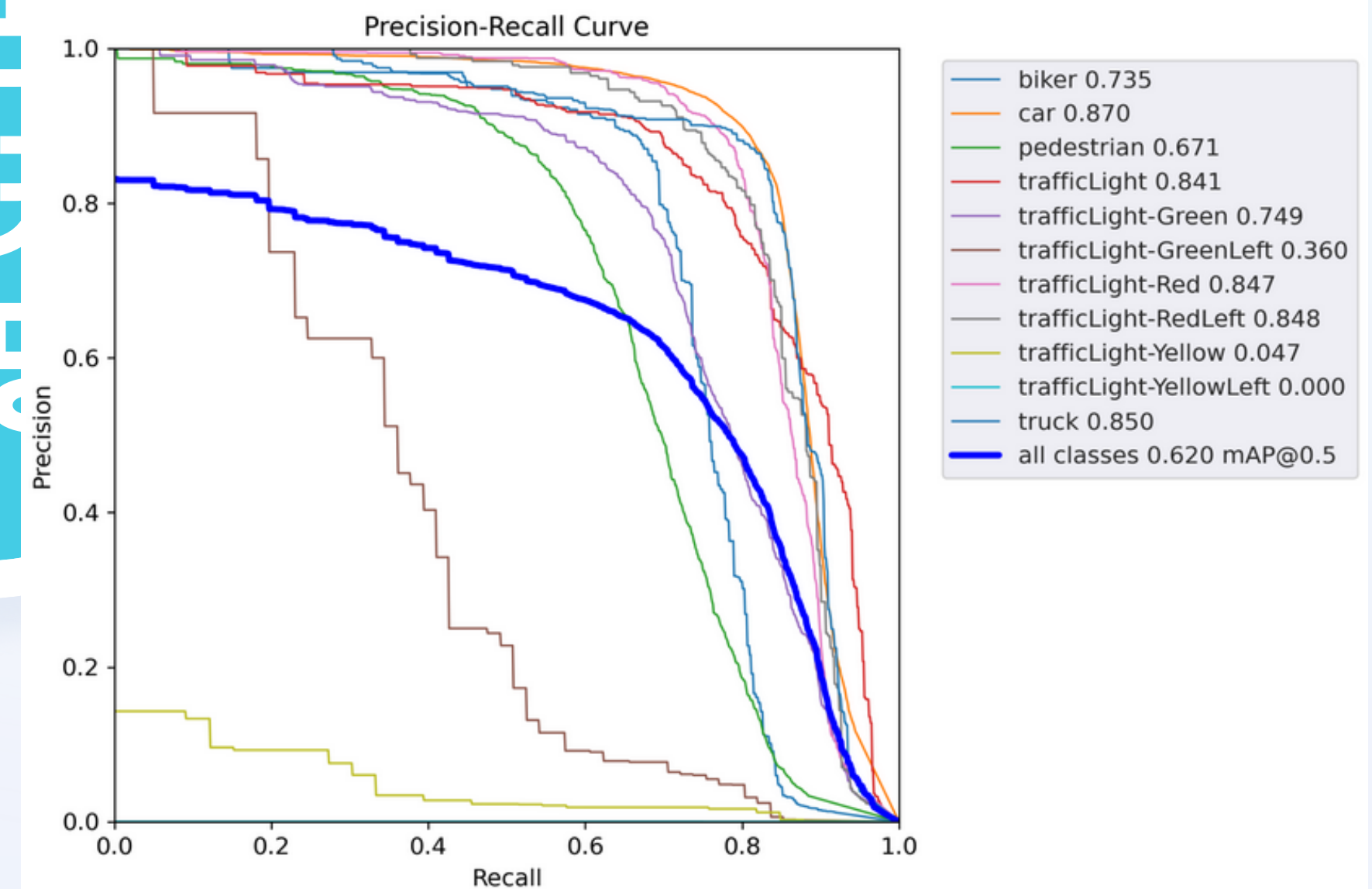
PR-curve

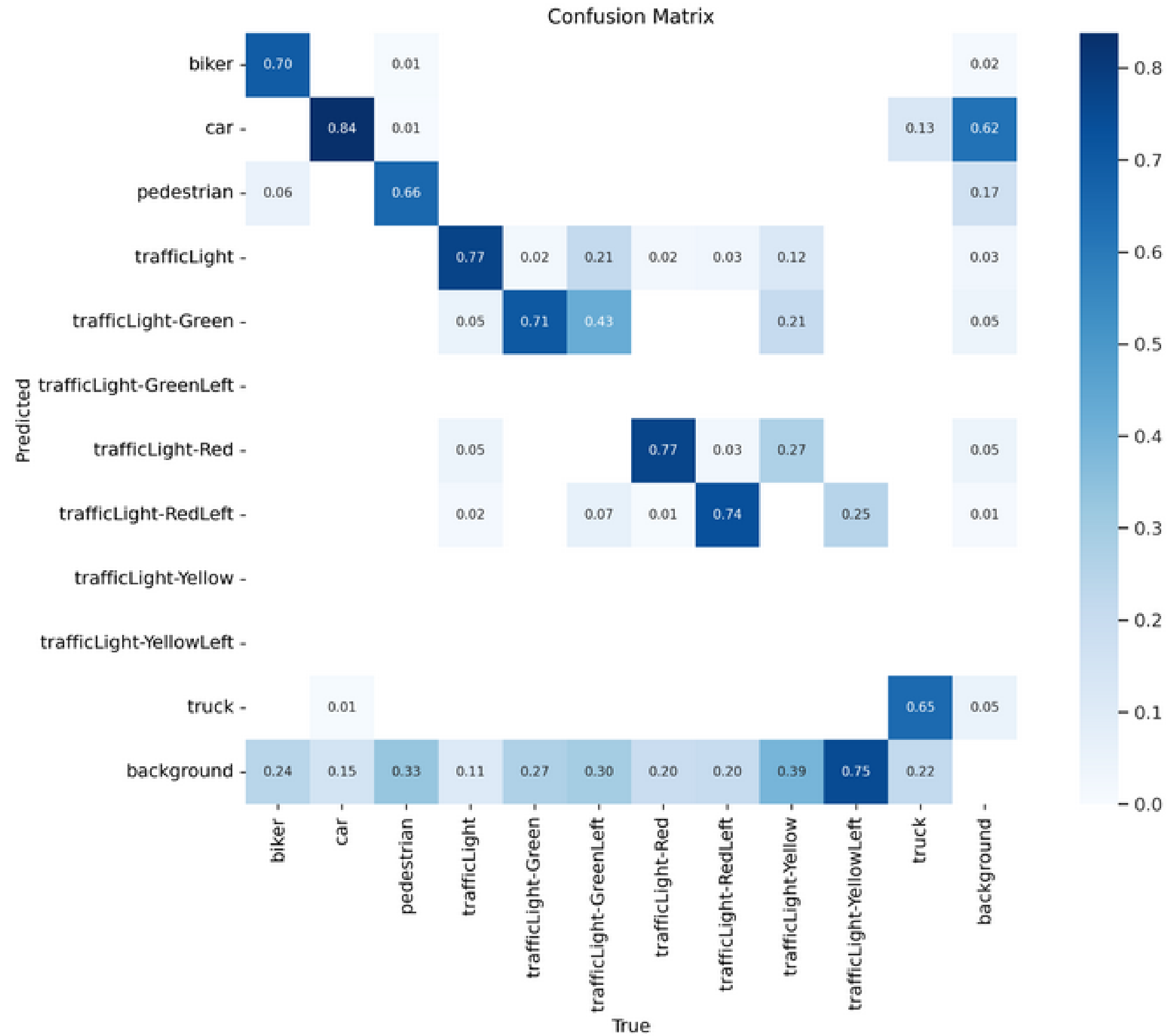
'car/Red/RedLeft'는 높은 P·R, 반면
Yellow/GreenLeft는 샘플 부족으로 곡선이 바닥.



아직 수렴하지 않은 것을 보아 epoch를 더 줘도 될 것 같음

Loss





10 epochs completed in 2.231 hours.

Optimizer stripped from runs/train/udacity_yolov5m_xpu3/weights/last.pt, 84.1MB

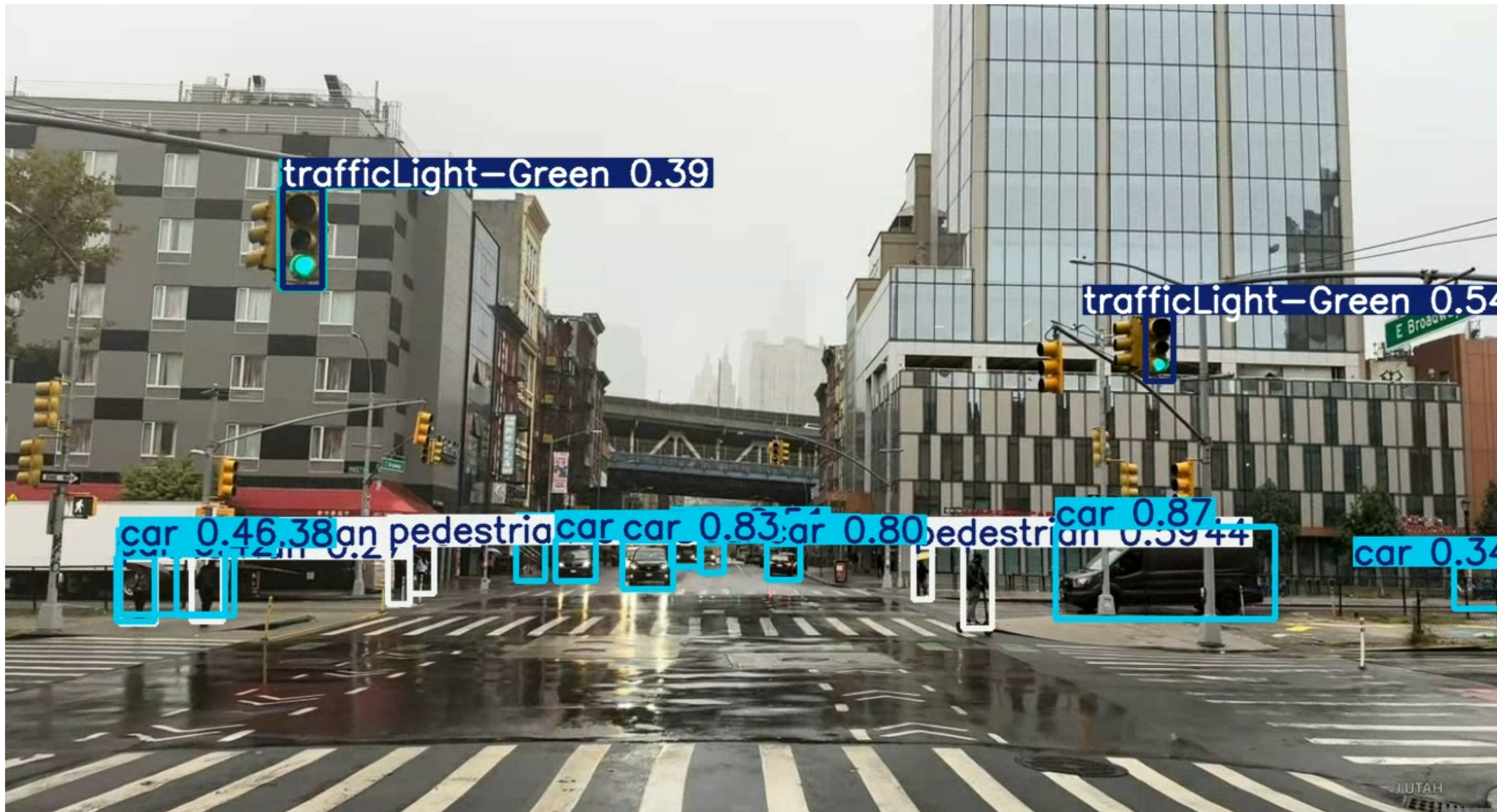
Optimizer stripped from runs/train/udacity_yolov5m_xpu3/weights/best.pt, 84.1MB

Validating runs/train/udacity_yolov5m_xpu3/weights/best.pt...

Fusing layers...

Model summary: 212 layers, 20893344 parameters, 0 gradients, 48.0 GFLOPs

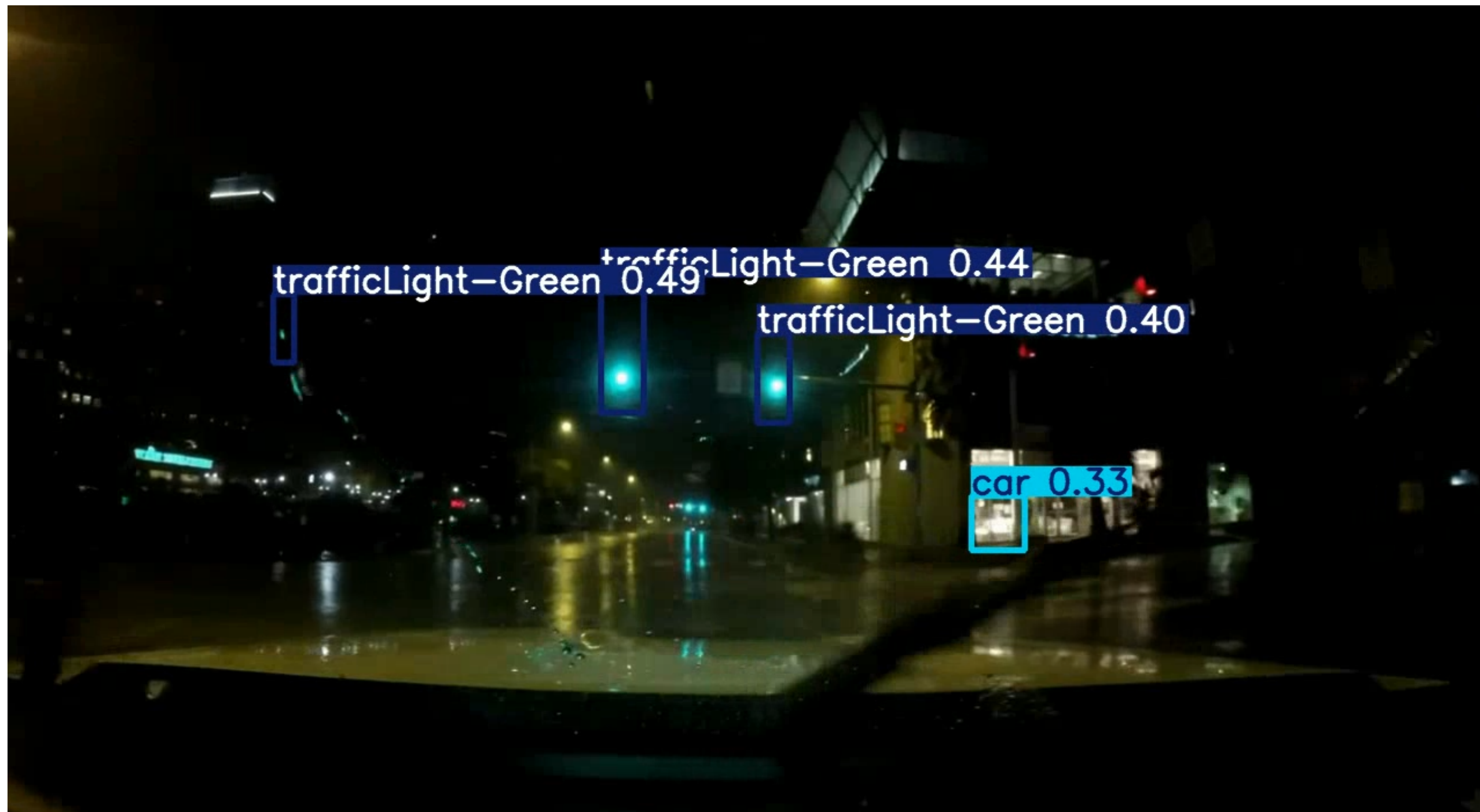
Class	Images	Instances	P	R	mAP50	mAP50-95:
all	2980	19480	0.889	0.532	0.62	0.336
biker	2980	356	0.854	0.689	0.735	0.372
car	2980	12870	0.87	0.82	0.87	0.562
pedestrian	2980	2019	0.773	0.596	0.671	0.31
trafficLight	2980	489	0.8	0.779	0.841	0.476
trafficLight-Green	2980	1130	0.811	0.66	0.749	0.306
trafficLight-GreenLeft	2980	61	1	0	0.36	0.187
trafficLight-Red	2980	1390	0.896	0.771	0.847	0.451
trafficLight-RedLeft	2980	353	0.894	0.74	0.848	0.444
trafficLight-Yellow	2980	33	1	0	0.047	0.0187
trafficLight-YellowLeft	2980	4	1	0	0.0002	5.54e-05
truck	2980	775	0.881	0.8	0.85	0.567





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JUTAH

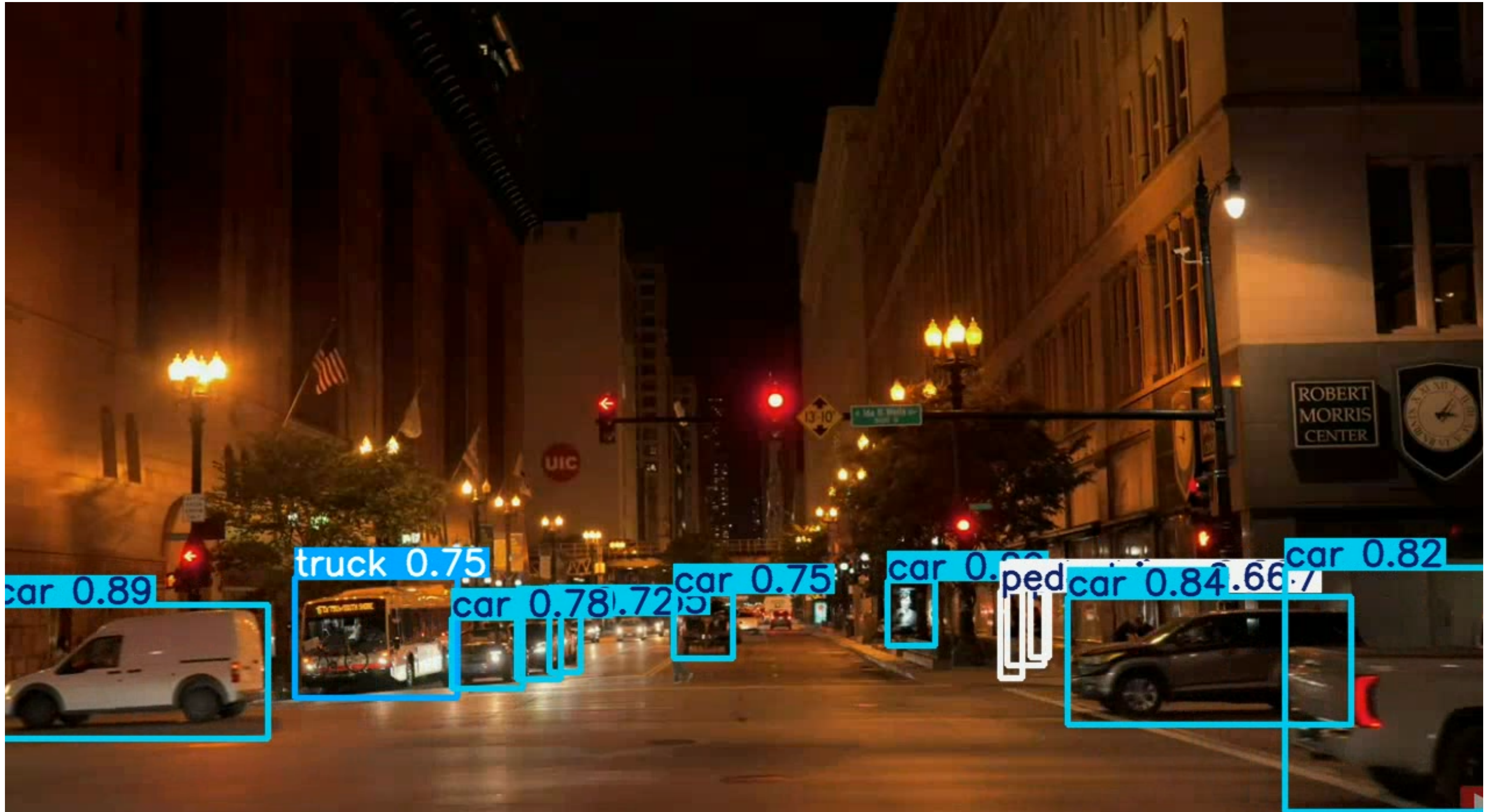


trafficLight-Green 0.49

trafficLight-Green 0.44

trafficLight-Green 0.40

car 0.33



car 0.89

truck 0.75

car 0.78

car 0.75

car 0.66

ped car 0.84

car 0.82